

PROVINCETOWN, MA, USA

WMO: 725073

Lat: 42.072N

Lon: 70.221W

Elev: 8

StdP: 14.69

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 64708

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	12.8	17.9	0.2	5.6	16.7	3.0	6.4	19.8	35.0	28.7	32.9	27.7	18.6	320	0.596

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.0	83.6	72.8	81.2	70.9	79.0	69.6	74.8	81.0	73.2	78.4	71.6	75.9	10.2	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
72.5	120.6	78.2	71.8	117.6	77.0	70.2	111.2	75.0	38.2	81.1	36.6	78.5	35.3	75.9	82.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
29.9	26.5	23.9	DB	7.9	89.3	4.6	3.0	4.6	91.4	1.9	93.2	-0.7	94.8	-4.0	97.0	
			WB	6.4	77.6	4.5	2.2	3.2	79.2	0.6	80.5	-1.9	81.8	-5.1	83.4	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	51.0	32.5	32.6	37.6	45.7	55.0	64.1	71.2	69.9	63.7	54.6	45.3	38.0
	DBStd	15.04	8.29	7.31	6.58	5.96	6.23	6.03	4.66	4.89	5.85	6.32	7.01	7.46
	HDD50	2167	542	488	388	154	20	0	0	0	0	28	171	376
	HDD65	5615	1007	908	851	578	317	87	7	14	93	326	590	837
	CDD50	2517	0	0	3	26	173	424	657	618	411	170	31	4
	CDD65	491	0	0	0	0	6	61	200	167	54	3	0	0
	CDH74	1966	0	0	0	5	39	237	893	663	125	4	0	0
	CDH80	341	0	0	0	1	6	40	183	99	12	0	0	0
Wind	WSAvg	11.5	14.4	13.9	13.2	11.4	10.1	9.5	8.9	8.5	9.6	11.7	13.1	14.1
Precipitation	PrecAvg	42.5	3.9	3.3	3.9	3.6	3.1	3.0	2.4	3.2	3.6	3.7	4.5	4.4
	PrecMax	57.5	12.9	9.1	9.9	9.4	8.1	7.6	5.3	7.3	9.6	9.1	9.4	8.9
	PrecMin	22.9	1.1	0.8	0.8	0.9	0.6	0.5	0.6	0.1	0.9	1.3	1.2	1.3
	PrecStd	9.0	2.9	1.7	2.1	1.8	1.8	2.0	1.3	1.9	2.1	2.0	2.2	1.8
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	53.8	52.6	60.4	68.4	79.5	84.4	88.5	85.9	81.4	72.8	63.4	57.3
		MCWB	52.3	49.5	51.1	58.6	64.7	71.2	74.7	75.1	71.5	66.0	58.4	53.9
	2%	DB	49.6	48.0	53.6	62.2	72.4	80.0	84.2	82.2	77.5	69.6	61.0	54.5
		MCWB	47.9	45.3	49.2	53.2	62.2	69.0	72.7	72.3	69.3	64.1	56.8	52.0
	5%	DB	46.2	45.4	49.4	57.2	68.2	76.4	81.6	80.8	74.5	66.0	57.4	52.3
		MCWB	43.8	42.3	45.5	50.5	59.0	66.3	71.2	71.0	67.6	62.2	53.5	50.0
	10%	DB	44.4	43.0	46.1	54.6	64.0	73.0	79.2	78.5	72.3	63.7	55.2	49.1
		MCWB	41.3	39.3	42.1	49.2	57.1	64.7	69.9	69.7	66.7	58.8	51.3	46.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	52.6	50.6	53.0	60.1	67.6	73.8	77.2	76.9	74.4	68.7	61.0	55.6
		MCDB	53.2	51.6	55.3	66.1	74.8	81.6	86.3	83.2	80.0	71.3	62.5	56.8
	2%	WB	47.8	46.1	49.1	55.8	64.0	71.0	75.0	74.8	71.4	65.3	57.9	52.9
		MCDB	48.8	47.4	51.9	59.9	69.6	77.5	81.6	80.3	74.7	67.0	60.6	54.1
	5%	WB	44.9	43.1	46.0	53.0	61.4	68.9	73.3	73.1	69.7	62.9	55.0	50.1
		MCDB	46.1	45.2	48.0	56.1	66.6	73.9	79.0	77.8	72.8	65.1	57.3	51.8
	10%	WB	41.4	39.8	42.8	50.1	58.7	66.7	71.7	71.5	67.5	60.1	52.6	46.7
		MCDB	43.9	42.4	45.8	54.1	63.4	71.2	76.4	75.2	70.9	63.1	55.1	48.4
Mean Daily Temperature Range	5% DB	MDBR	10.3	11.3	11.0	12.7	14.1	14.2	14.0	13.7	13.1	11.9	11.3	10.4
		MCDBR	12.9	14.1	15.4	17.8	20.4	18.5	16.8	15.6	14.8	14.2	13.2	13.5
		MCWBR	13.1	13.1	11.9	11.3	11.4	9.7	8.0	8.2	9.1	10.5	12.2	14.3
	5% WB	MCDBR	13.2	14.6	14.5	15.5	17.7	15.9	14.5	13.5	12.5	12.9	12.5	13.6
		MCWBR	14.1	14.6	13.0	12.1	11.6	9.9	8.2	8.2	8.8	11.3	13.0	14.8
Clear-Sky Solar Irradiance	taub		0.306	0.312	0.330	0.371	0.406	0.458	0.480	0.439	0.376	0.343	0.317	0.303
	taud		2.520	2.509	2.498	2.403	2.354	2.240	2.185	2.292	2.448	2.539	2.572	2.572
	Ebn at Noon		269	285	292	285	276	261	253	261	272	270	263	261
	Edh at Noon		23	27	30	36	39	43	45	40	32	26	22	20
All-Sky Solar Radiation	RadAvg		530	816	1153	1505	1701	1828	1902	1712	1379	928	607	448
	RadStd		40	75	86	161	170	165	111	77	93	77	46	49

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (58 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+169	+77

Nomenclature: See separate page